Table 1: End-to-end execution time (s, median per iteration) and speedup split by pass type. ADT fields = non-recursive fields annotated with `@BENCH adt_fields`; SoA bufs = 1+ non-recursive field slots across all constructors (recursive children are stored in the tag buffer). Speedup >1× means SoA is faster; **bold** marks >1.1×.

Program	**ADT**	**SoA**	**Fold passes**			**Map passes**		
	fields	bufs	AoS (s)	SoA (s)	Speedup	AoS (s)	SoA (s)	Speedup
KDTree	17	16	0.1349	0.1011	**1.33**×	–	–	–

Table 2: PAPI speedup summary across all passes. For each program and counter, speedup is AoS/SoA using summed per-pass median counters. >1× means SoA has fewer events.

Program	**CYC**	**INS**	**L1D**	**L1I**	**L2D**	**L2I**	**LLC**
KDTree	**1.33**×	0.78×	**1.35**×	0.84×	0.23×	0.85×	**5.21**×
Geomean	**1.33**×	0.78×	**1.35**×	0.84×	0.23×	0.85×	**5.21**×

Table 3: PAPI counter totals across all passes (AoS/SoA). Each entry is the sum of per-pass median counter values.

Program	**CYC (A/S)**	**INS (A/S)**	**L1D (A/S)**	**L1I (A/S)**	**L2D (A/S)**	**L2I (A/S)**	**LLC (A/S)**
KDTree	6.8e+08/**5.1e+08**	**2.3e+09**/3.0e+09	1.4e+07/**1.0e+07**	**6.7e+02**/8.0e+02	**1.9e+04**/8.1e+04	**6.8e+02**/8.0e+02	3.9e+07/**7.4e+06**

Table 4: Per-pass performance for `KDTree`, ADT has 17 fields, SoA uses 16 buffers. Times are median per iteration (s); ± shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup >1× means SoA is faster. OOM = out of memory.

Pass	**T**	**Uses**	**Dead%**	**AoS med±err**	**SoA med±err**	**Speedup**
Find nearest Neighbour	F	13/17	24%	0.0270±0.00045	**0.0212±0.00024**	**1.28**×
countInRange tight_box	F	13/17	24%	0.0277±0.00003	**0.0215±0.00002**	**1.29**×
pointCloudNeighborhood	F	11/17	35%	0.0246±0.00003	**0.0162±0.00001**	**1.52**×
sumMassInRange	F	14/17	18%	0.0275±0.00130	**0.0222±0.00002**	**1.24**×
twoPointCorrelation bin_8_16	F	11/17	35%	0.0282±0.00718	**0.0201±0.00004**	**1.40**×
Total				0.1349	0.1011	**1.33**×
Geomean						**1.34**×

Table 5: Per-pass PAPI counters for `KDTree` (A/S). Each cell is median counter value pair per pass. Counter headers are abbreviated for compactness: CYC=CPU cycles, L1D/L1I/L2D/L2I=cache misses, LLC=LLC load misses.

Pass	**T**	**CYC (A/S)**	**INS (A/S)**	**L1D (A/S)**	**L1I (A/S)**	**L2D (A/S)**	**L2I (A/S)**	**LLC (A/S)**
Find nearest Neighbour	F	1.4e+08/**1.1e+08**	**5.1e+08**/6.4e+08	3.2e+06/**2.0e+06**	2.3e+02/**1.4e+02**	**3.5e+03**/1.6e+04	2.3e+02/**1.4e+02**	7.8e+06/**1.6e+06**
countInRange tight_box	F	1.4e+08/**1.1e+08**	**4.6e+08**/6.0e+08	3.1e+06/**2.1e+06**	**7.8e+01**/3.2e+02	**4.2e+03**/1.6e+04	**8.1e+01**/3.2e+02	7.8e+06/**1.6e+06**
pointCloudNeighborhood	F	1.2e+08/**8.2e+07**	**3.5e+08**/4.7e+08	2.7e+06/**2.3e+06**	1.4e+02/**7.1e+01**	**3.9e+03**/1.6e+04	1.4e+02/**7.1e+01**	7.8e+06/**1.3e+06**
sumMassInRange	F	1.4e+08/**1.1e+08**	**4.6e+08**/6.2e+08	**3.0e+06**/3.3e+06	**1.2e+02**/1.4e+02	**4.2e+03**/2.0e+04	**1.2e+02**/1.4e+02	7.6e+06/**1.7e+06**
twoPointCorrelation bin_8_16	F	1.4e+08/**1.0e+08**	**5.1e+08**/6.1e+08	2.2e+06/**7.1e+05**	**1.1e+02**/1.3e+02	**3.1e+03**/1.3e+04	**1.1e+02**/1.3e+02	7.7e+06/**1.3e+06**
Total		6.8e+08/5.1e+08	2.3e+09/3.0e+09	1.4e+07/1.0e+07	6.7e+02/8.0e+02	1.9e+04/8.1e+04	6.8e+02/8.0e+02	3.9e+07/7.4e+06